PHOTOLITHOGRAPHY PROCESS IN VLSI

Pattern Transfer Technique in Semiconductor Fabrication

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Introduction

Photolithography Definition:

- Photolithography is a process used to transfer circuit patterns onto a silicon wafer using UV light.
- It is one of the most important steps in semiconductor manufacturing.

Purpose:

- To create micro-scale patterns for integrated circuits.

Overall Process

Main Steps Involved:

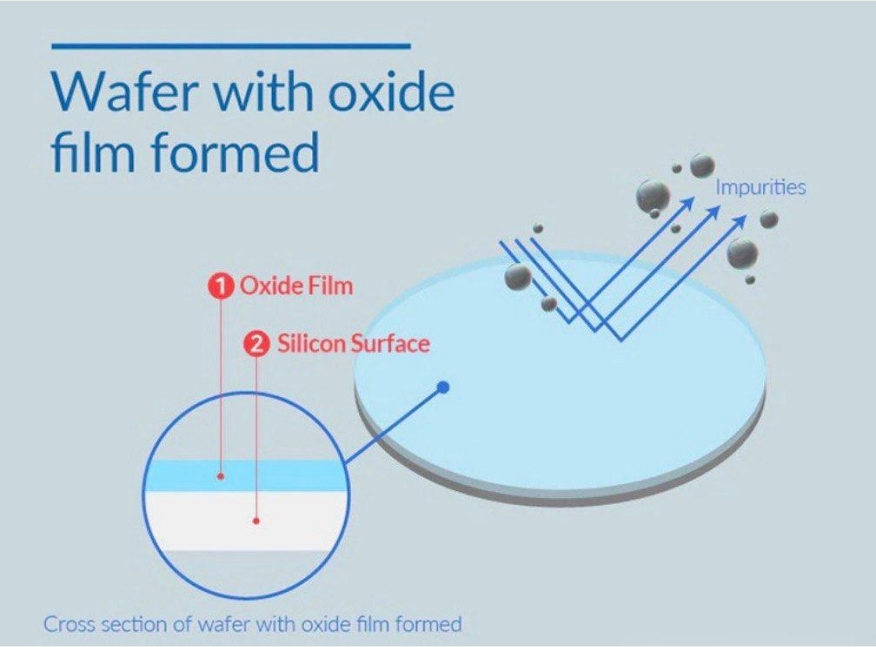
- Wafer Cleaning
- Oxide Layer Formation
- Photoresist Coating
- Soft Baking
- Mask Alignment
- UV Exposure
- Development
- Hard Baking
- Etching
- Photoresist Removal

Wafer Cleaning

- The silicon wafer is cleaned to remove:
  - Dust
  - Organic materials
  - Contaminants
- Ensures proper adhesion of layers
- Common method: RCA cleaning

Oxide Layer Formation

- A thin layer of Silicon Dioxide (SiO_2) is grown
- Done using thermal oxidation
- Acts as:
  - Insulating layer
  - Protective layer

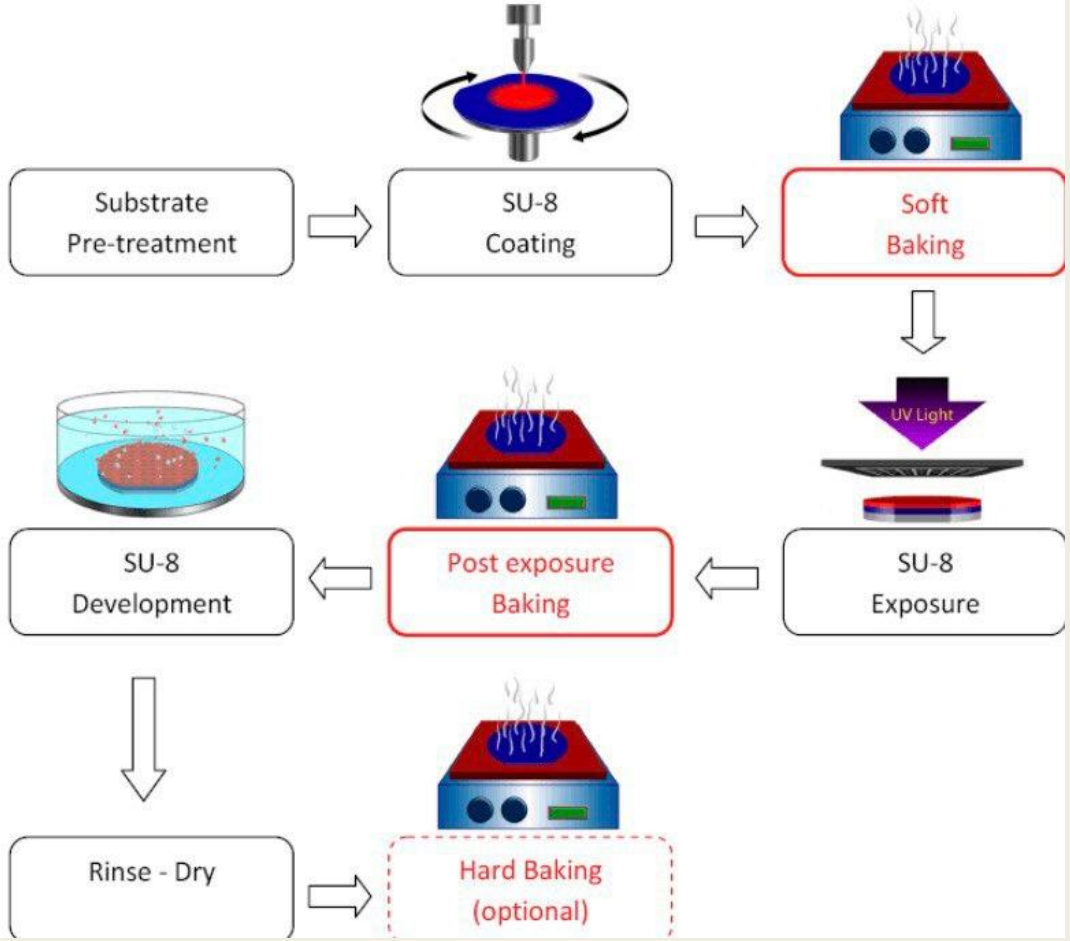


Photoresist Coating

- A light-sensitive material is applied
- Spin coating is used for uniform thickness
- Two types:
  - Positive photoresist
  - Negative photoresist

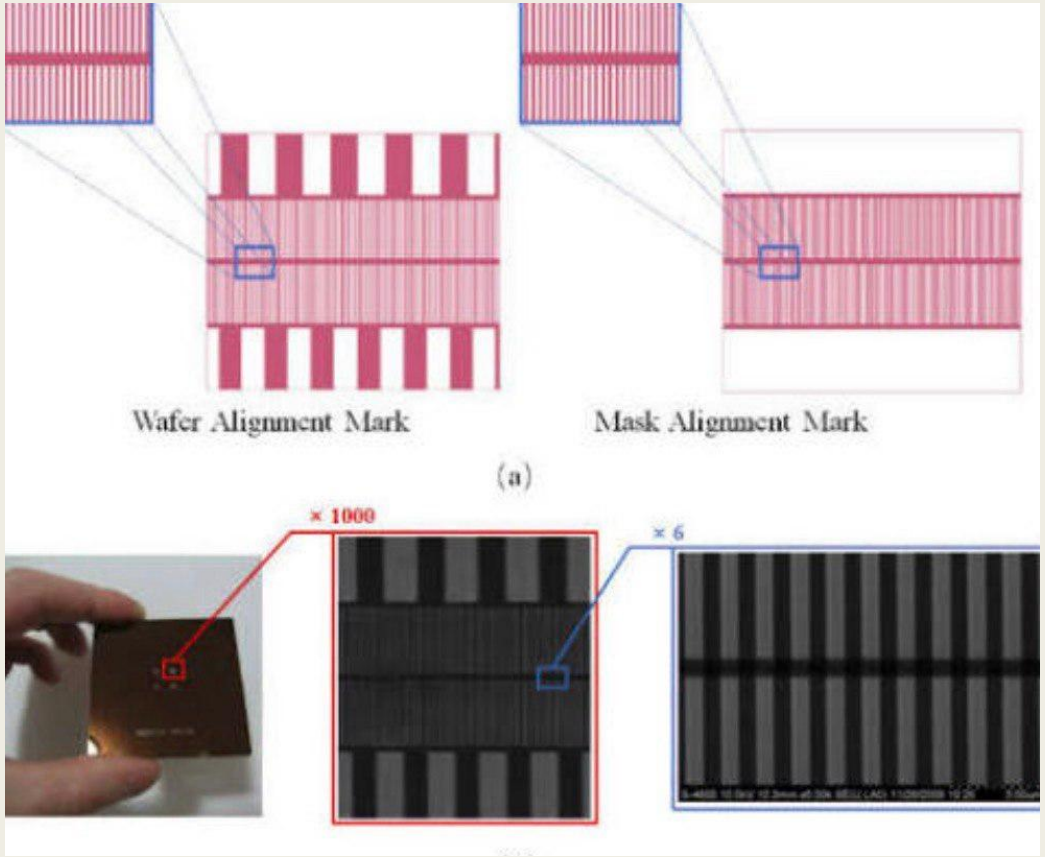
Soft Baking

- Heated at 90–100°C
- Removes excess solvent
- Improves adhesion and stability



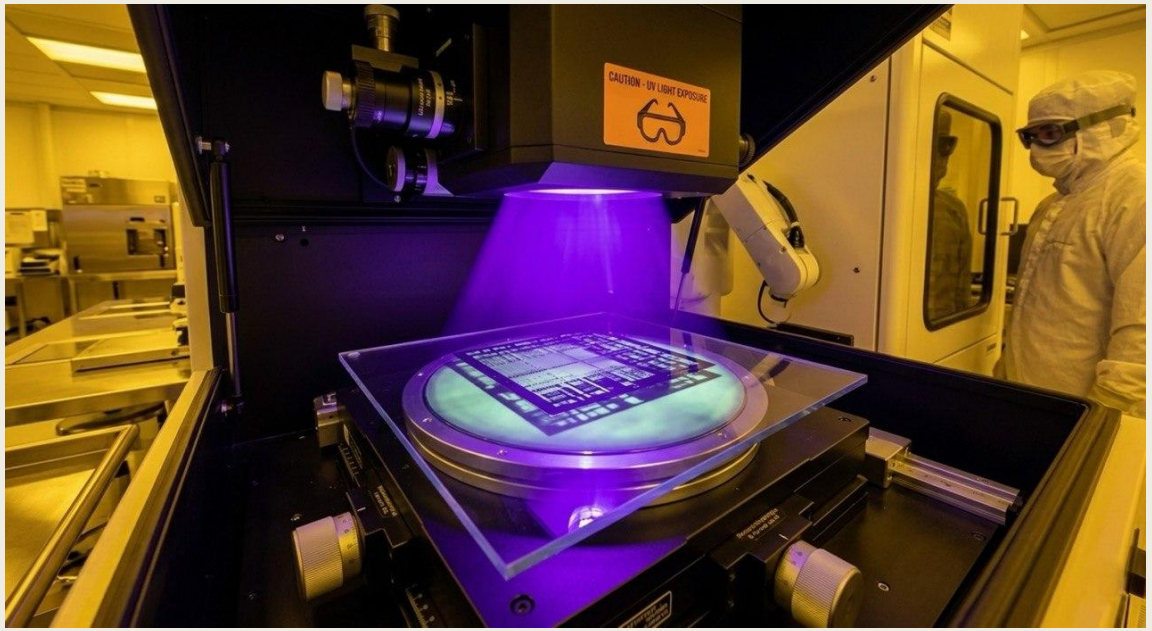
Mask Alignment

- A photomask containing the circuit pattern is aligned
- Requires high precision
- Ensures accurate pattern transfer



UV Exposure

- Ultraviolet light is passed through the mask
- Causes chemical changes in photoresist
- Defines the pattern area



Development

- Wafer is placed in developer solution

Types:

- Positive resist → exposed areas removed
- Negative resist → unexposed areas removed

Hard Baking

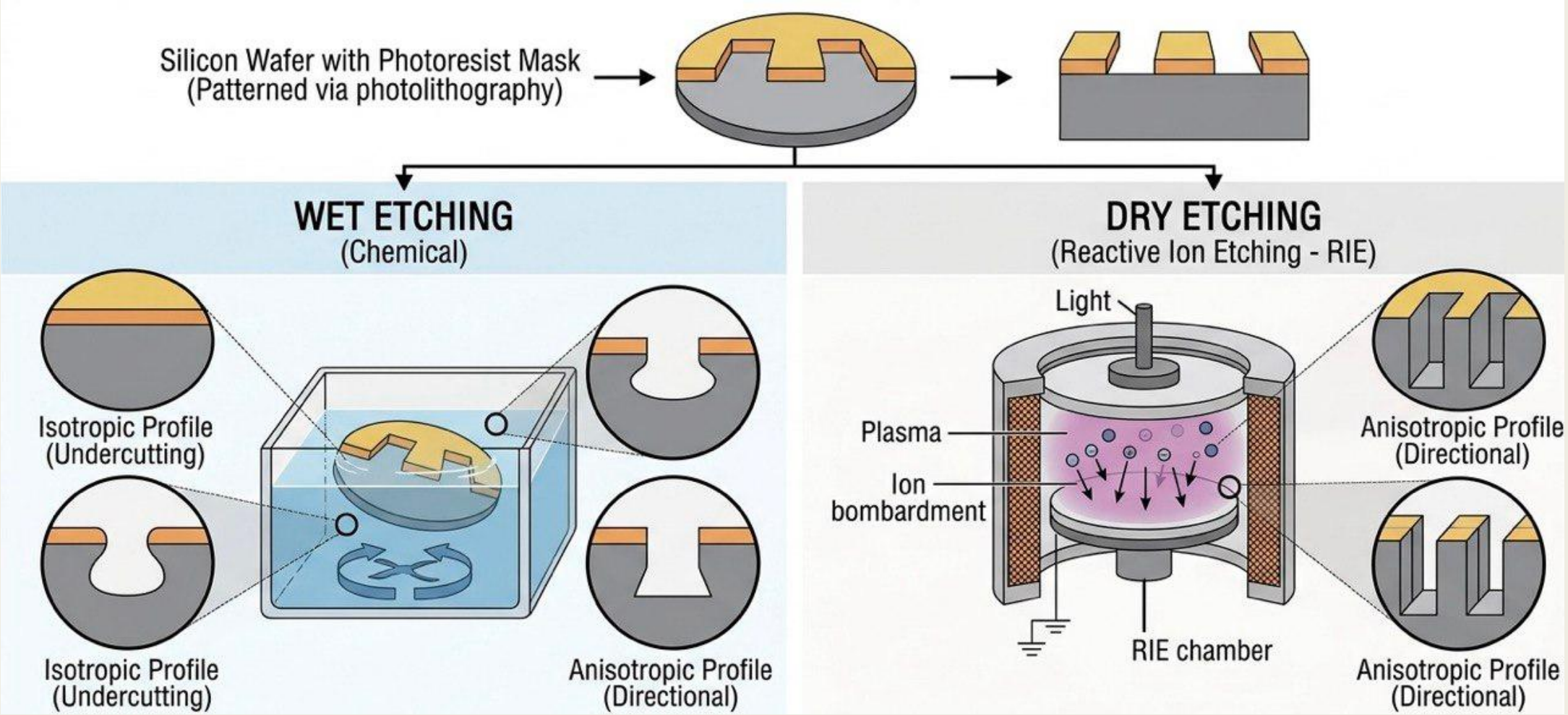
- Heated again to harden photoresist
- Improves resistance to etching

Etching

- Removes exposed oxide layer
- Transfers pattern onto wafer

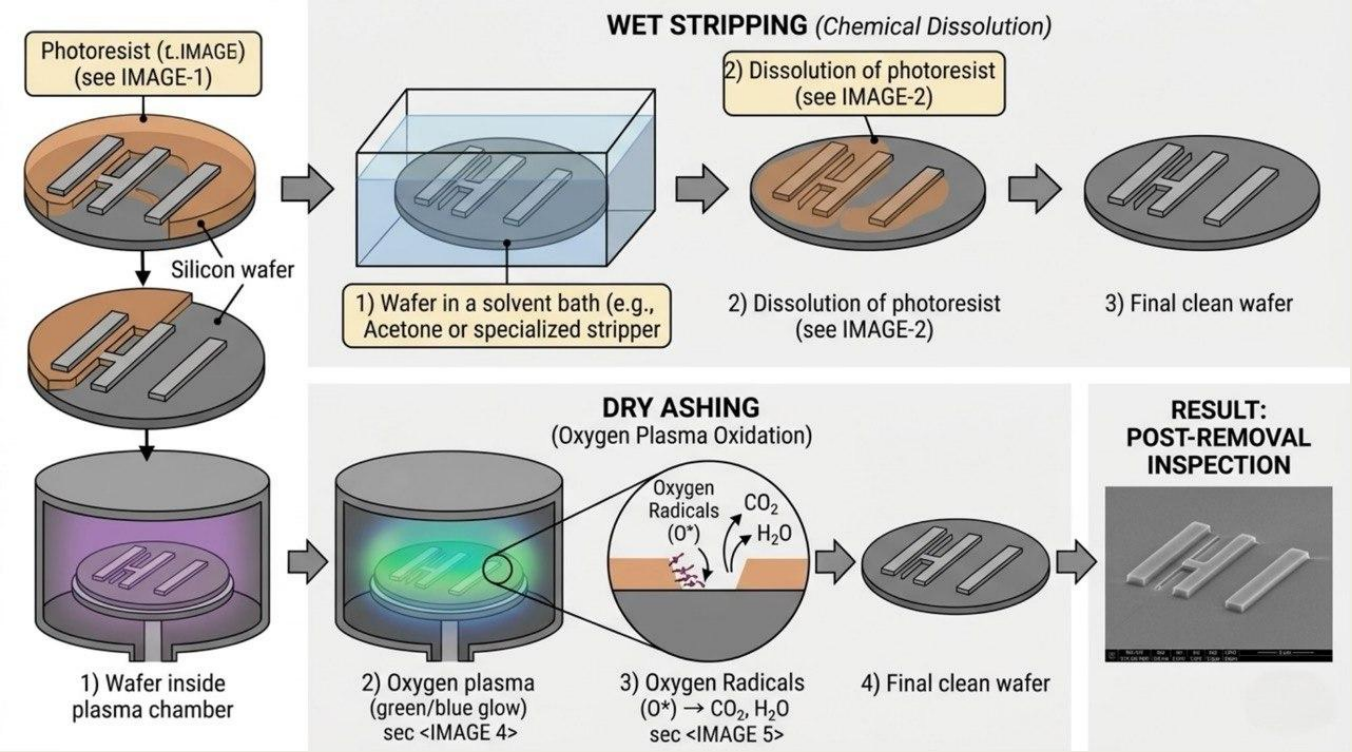
Types:

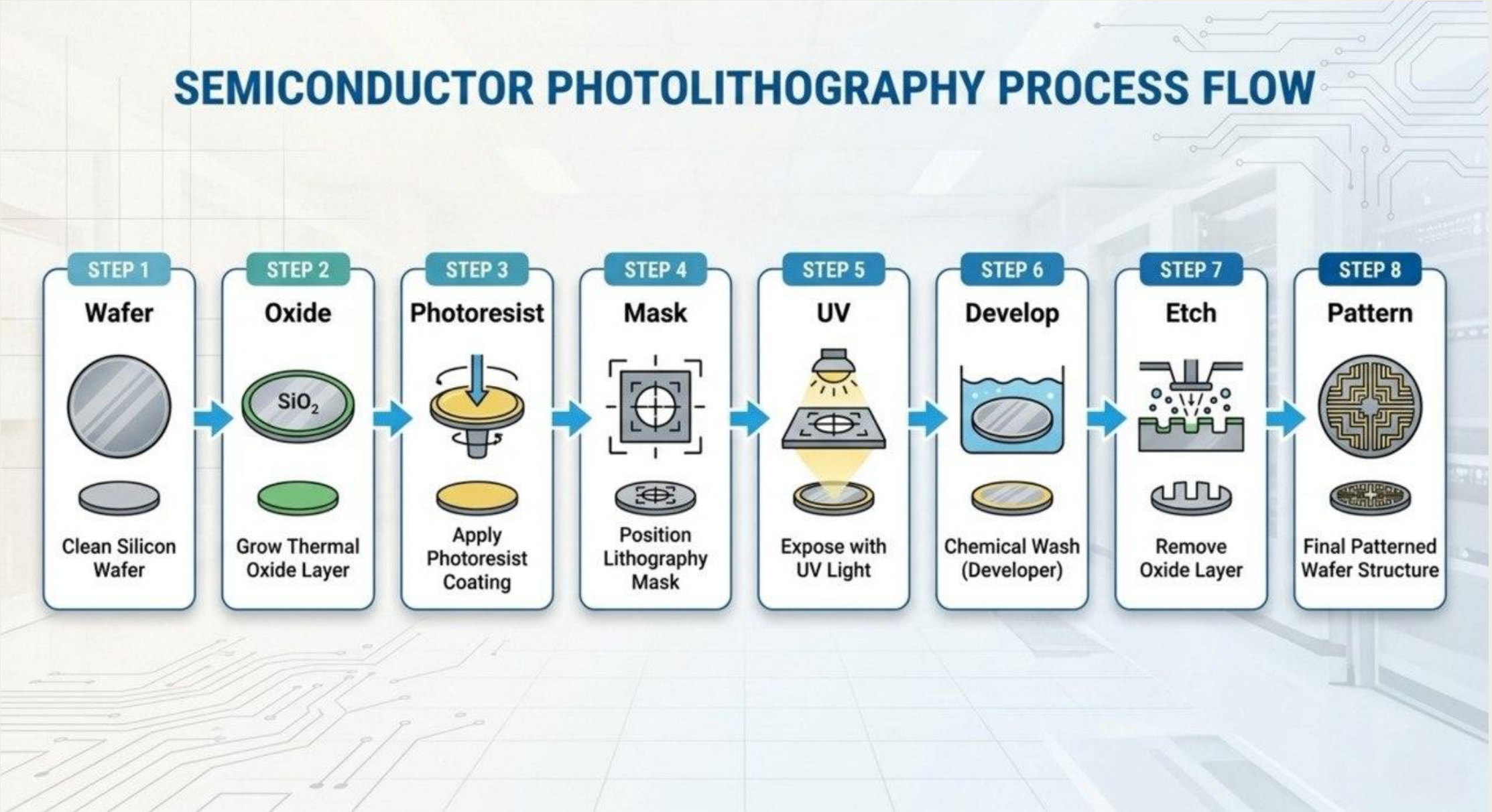
- Wet etching
- Dry etching

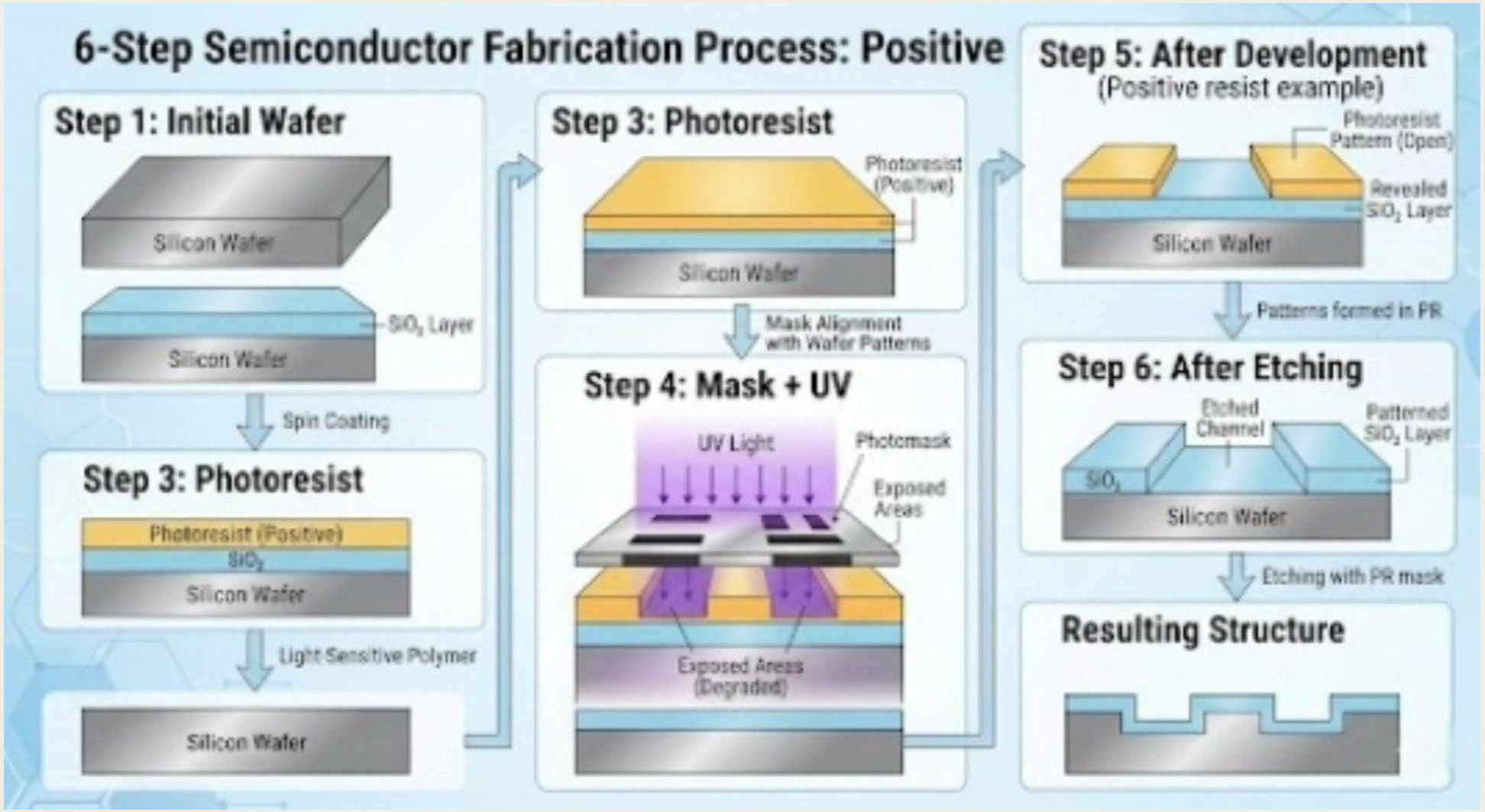

Silicon Wafer with Photoresist Mask
(Patterned via photolithography)
WET ETCHING
(Chemical)
Isotropic Profile
(Undercutting)
Isotropic Profile
(Undercutting)
Anisotropic Profile
(Directional)
DRY ETCHING
(Reactive Ion Etching - RIE)
Light
Plasma
Ion
bombardment
RIE chamber
Anisotropic Profile
(Directional)
Anisotropic Profile
(Directional)

Photoresist Removal

- Remaining photoresist is removed
- Final patterned layer remains




SEMICONDUCTOR PHOTOLITHOGRAPHY PROCESS FLOW
STEP 1
Wafer
Clean Silicon Wafer
STEP 2
Oxide
SiO_2
Grow Thermal Oxide Layer
STEP 3
Photoresist
Apply Photoresist Coating
STEP 4
Mask
Position Lithography Mask
STEP 5
UV
Expose with UV Light
STEP 6
Develop
Chemical Wash (Developer)
STEP 7
Etch
Remove Oxide Layer
STEP 8
Pattern
Final Patterned Wafer Structure


6-Step Semiconductor Fabrication Process: Positive
Step 1: Initial Wafer
Silicon Wafer
SiO_2 Layer
Silicon Wafer
Spin Coating
Step 3: Photoresist
Photoresist (Positive)
SiO_2
Silicon Wafer
Light Sensitive Polymer
Silicon Wafer
Step 3: Photoresist
Photoresist (Positive)
Silicon Wafer
Mask Alignment with Wafer Patterns
Step 4: Mask + UV
UV Light
Photomask
Exposed Areas
Exposed Areas (Degraded)
Step 5: After Development
(Positive resist example)
Photoresist Pattern (Open)
Revealed SiO_2 Layer
Silicon Wafer
Patterns formed in PR
Step 6: After Etching
Etched Channel
Patterned SiO_2 Layer
SiO_2
Silicon Wafer
Etching with PR mask
Resulting Structure

Final Result

- The desired circuit pattern is successfully transferred onto the wafer surface
- Used in IC fabrication

Conclusion

- Photolithography is a key process in VLSI
- Enables miniaturization of circuits
- Essential for modern electronics